

RESEARCH

Open Access



Soil erosion risk assessment using RUSLE and geospatial techniques in Peruvamba sub-watershed, Kerala, India

Sapna Kinattinkara¹, Vinaykumar Gaddam^{2*} , Thangavelu Arumugam³, Karpagamani Vellingiri⁴ and Manimaran Arumugam⁵

*Correspondence:

Vinaykumar Gaddam

gaddam_vinay@gmail.com

¹Department of Environmental Science, PSG College of Arts and Science, Coimbatore 641014, India

²Department of Civil Engineering, Siddhartha Academy of Higher Education, Vijayawada, Andhra Pradesh 521137, India

³Department of Environmental Studies, Kannur University, Kannur, Kerala 670567, India

⁴Department of Mathematics, Karpagam Institute of Technology, Coimbatore 641105, India

⁵Department of Mathematics, Kongu Engineering College, Erode, Tamil Nadu 638060, India

Abstract

Soil erosion is a major environmental challenge in the humid tropical landscapes of the Western Ghats, where intense monsoonal rainfall, steep terrain, and land-use changes accelerate land degradation. This study evaluates soil erosion risk in the Peruvamba sub-watershed of Kerala, India, using the Revised Universal Soil Loss Equation (RUSLE) integrated with Geographic Information System (GIS) techniques. Rainfall erosivity (R), soil erodibility (K), slope length and steepness (LS), cover management (C), and conservation practice (P) factors were derived from multi-year rainfall records (2014–2023), soil survey data, satellite imagery, and digital elevation models. The average rainfall erosivity was estimated at 2710.67 MJ mm ha⁻¹ h⁻¹ yr⁻¹, while the LS-factor reached a maximum value of 28.88 in steep upland areas, indicating a strong topographic influence on erosion processes. The RUSLE-based assessment estimated a total annual soil loss of approximately 72,500 t yr⁻¹ and a mean soil loss of 17.4 t ha⁻¹ yr⁻¹, exceeding the commonly accepted soil-loss tolerance threshold for tropical regions. Forested and wetland areas exhibited low erosion rates, whereas agricultural lands, plantations, and scrub-dominated areas experienced moderate to severe erosion. Approximately 5.3% of the watershed was classified as very high erosion risk, while 14.7% of the area fell within high to very high erosion categories, representing priority zones for conservation intervention. The results demonstrate that topography and vegetation cover are the dominant controls on soil-loss variability and highlight the importance of targeted soil and water conservation measures, including contour bunding, terracing, and vegetation restoration. Comparison with published erosion estimates from comparable Western Ghats watersheds and sensitivity analysis indicated good agreement and supported the reliability of the model outputs. The integration of RUSLE and GIS provides a robust and scalable framework for erosion-risk assessment and supports sustainable watershed management in the Western Ghats and other tropical environments.

Keywords Soil erosion, RUSLE, GIS, Watershed management, Western Ghats, Kerala, Conservation prioritization



1 Introduction

Soil is a fundamental component of terrestrial ecosystems, supporting agricultural productivity, biodiversity conservation, water regulation, nutrient cycling, and carbon sequestration [1–5]. Through its role in storing organic carbon and regulating biogeochemical processes, soil also contributes significantly to climate regulation and ecosystem sustainability [6–15]. However, soil resources worldwide are increasingly threatened by degradation processes, among which soil erosion remains one of the most severe and widespread forms of land degradation [16–26]. Soil erosion involves the detachment, transport, and deposition of soil particles by rainfall and surface runoff, resulting in the loss of nutrient-rich topsoil and deterioration of land productivity [27–35]. The consequences of accelerated erosion extend beyond on-site land degradation and include sedimentation of rivers and reservoirs, declining water quality, increased flood risk, and disruption of ecosystem services [37–41]. The occurrence and severity of soil erosion are controlled by a combination of natural and anthropogenic factors [42]. Natural controls include rainfall intensity, topography, soil characteristics, and vegetation cover, whereas human activities such as deforestation, agricultural expansion, infrastructure development, mining, and unsustainable land-management practices often accelerate erosion rates beyond natural thresholds [43–45]. In tropical regions, particularly in monsoon-dominated countries such as India, the interaction between intense rainfall and rapid land-use change significantly increases erosion susceptibility [46]. Over time, accelerated soil loss reduces soil fertility, lowers agricultural productivity, and contributes to long-term environmental degradation [47]. Consequently, effective soil and water conservation measures are essential for maintaining ecosystem integrity and ensuring sustainable land management [48]. Practices such as afforestation, agroforestry, contour farming, terracing, conservation tillage, and watershed-based land-use planning have proven effective in reducing soil loss and restoring degraded landscapes [49–54]. However, the success of these measures depends on accurately identifying erosion-prone areas and understanding the relative influence of the factors controlling erosion processes [55, 56].

To support erosion assessment and conservation planning, numerous empirical and process-based models have been developed [57, 58]. Among these, the Revised Universal Soil Loss Equation (RUSLE) is one of the most widely applied approaches because of its conceptual simplicity, moderate data requirements, and adaptability to different environmental conditions [59]. RUSLE estimates long-term average annual soil loss by integrating rainfall erosivity (R), soil erodibility (K), slope length and steepness (LS), cover management (C), and conservation practice (P) factors [60, 61]. Despite its empirical nature, the model has demonstrated reliable performance across diverse climatic and physiographic settings and has been extensively applied in agricultural, forested, and mountainous environments [62–65]. The integration of RUSLE with Geographic Information Systems (GIS) and remote sensing has further enhanced its utility by enabling efficient processing of spatial datasets related to rainfall, topography, soils, and land use. This integrated approach facilitates the generation of erosion-risk maps, identification of erosion hotspots, and evaluation of land-use impacts at watershed and sub-watershed scales [66–71]. India experiences substantial soil erosion due to its dependence on agriculture, complex terrain, and exposure to intense monsoonal rainfall [69, 70]. The Western Ghats, recognized globally as a biodiversity hotspot, are particularly vulnerable because of their steep slopes, highly weathered lateritic soils, intense rainfall, and

increasing anthropogenic pressures [71–73]. Previous studies conducted in Kerala and other parts of the Western Ghats have reported considerable spatial variability in erosion rates and highlighted the importance of topography, vegetation cover, and rainfall intensity as major controlling factors. However, most investigations have focused on larger river basins or relied on short-term rainfall records and generalized land-use datasets [73–76]. Consequently, detailed information on erosion hotspots, factor interactions, and conservation priorities at the sub-watershed scale remains limited, particularly for the Peruvamba sub-watershed in Kannur District, Kerala.

The Peruvamba sub-watershed is characterized by undulating to steep terrain, high annual rainfall, heterogeneous land-use patterns, and increasing developmental pressures, making it highly susceptible to soil erosion. Despite its ecological and hydrological significance within the Western Ghats region, comprehensive spatially explicit assessments of soil erosion risk and conservation priorities remain scarce. Furthermore, projected increases in rainfall intensity associated with climate change may further exacerbate erosion processes in the region, highlighting the need for robust spatial assessments capable of supporting adaptive watershed management strategies. Against this background, the present study applies a GIS-integrated RUSLE framework to evaluate soil erosion dynamics in the Peruvamba sub-watershed. Unlike previous studies that primarily focused on basin-scale assessments, the present research combines a decade of rainfall observations (2014–2023), DEM-derived terrain analysis, NDVI-based vegetation assessment, uncertainty evaluation, and GIS-integrated RUSLE modelling to provide a comprehensive sub-watershed-scale erosion-risk assessment. The study further identifies conservation-priority zones to support evidence-based watershed management and climate-resilient land-use planning. To the best of our knowledge, this represents the first GIS-integrated RUSLE assessment incorporating a decade of rainfall observations (2014–2023) for conservation prioritization in the Peruvamba sub-watershed.

The objectives of this study were to:

- (i) Estimate spatial patterns of annual soil loss using a GIS-integrated RUSLE framework;
- (ii) Identify erosion hotspots and classify erosion-risk zones within the Peruvamba sub-watershed;
- (iii) Evaluate the relative influence of major RUSLE factors on erosion variability; and
- (iv) Develop conservation-priority recommendations to support sustainable watershed management.

2 Materials and methods

2.1 Study area

The Peruvamba sub-watershed is situated in the northern part of Kannur district, Kerala, India, and covers an area of approximately 269 km². Geographically, the watershed extends between latitudes 12°03'N to 12°15'N and longitudes 75°12'E to 75°24'E. Elevation within the watershed ranges from low-lying plains to a maximum height of about 1,145 m above mean sea level, reflecting pronounced topographic variability. The drainage system is dominated by the Peruvamba River, which is supported by several tributaries including Macharuthodu, Mathamangalam, Challachal, Mukkutenkara-chal, Nitarinapuzha, and Panappuzha. Geologically, the watershed comprises diverse

lithological units such as pyroxene granulites, biotite–hornblende gneiss, anthracite formations, coastal alluvium, sandstone, and clay deposits. The location and physiographic characteristics of the Peruvamba sub-watershed are shown in Fig. 1.

The upper catchment areas are predominantly characterised by lateritic soils, which are moderately erodible and highly sensitive to intense rainfall events. The watershed forms part of the ecologically fragile Western Ghats region and plays a crucial role in regional hydrology, agriculture, and ecosystem services. Ongoing land-use changes, agricultural expansion, and infrastructure development have increased the susceptibility of this landscape to soil erosion.

2.2 Data sources and preparation

Multiple spatial and non-spatial datasets were utilized to estimate soil erosion risk in the Peruvamba sub-watershed. Monthly rainfall data for the period 2014–2023 were obtained from the Indian Meteorological Department (IMD) and used for rainfall erosivity estimation. Soil texture information was sourced from the Soil Survey Organization of Kerala and served as the basis for determining soil erodibility. A Shuttle Radar

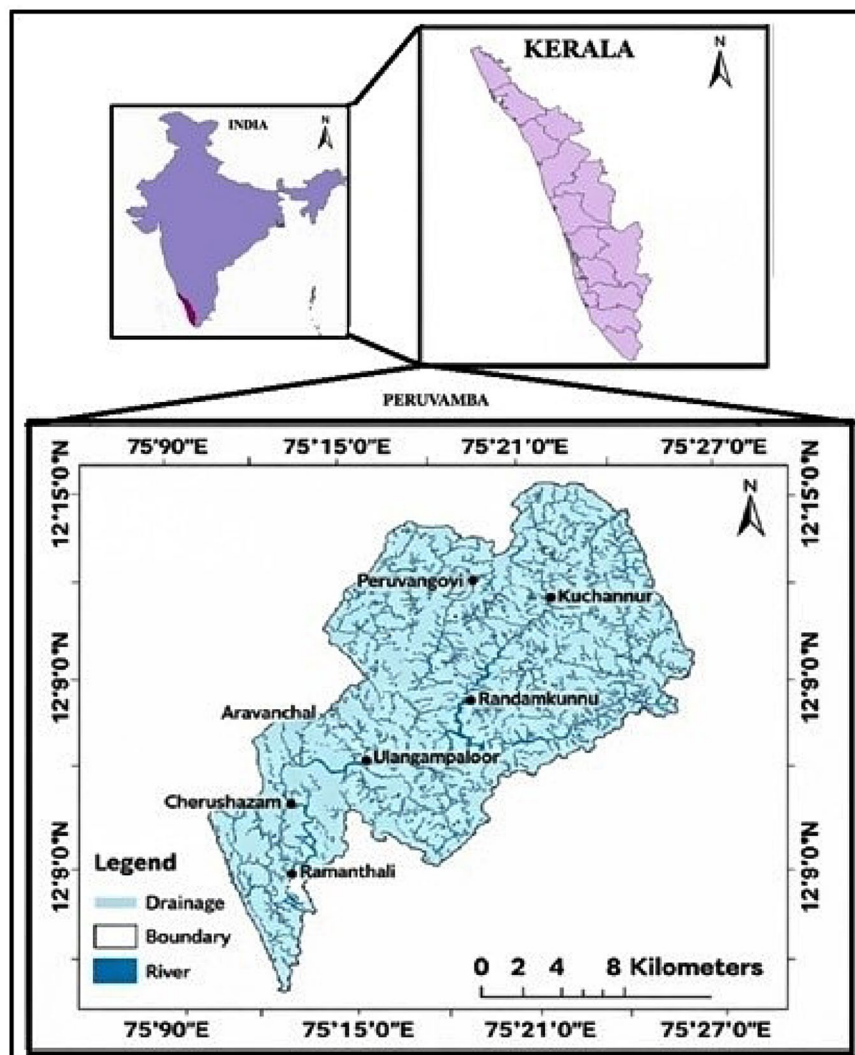


Fig. 1 Study area map of Peruvamba watershed

Topography Mission (SRTM) Digital Elevation Model (DEM) with 30 m spatial resolution was used to derive topographic parameters including slope and flow accumulation required for LS-factor estimation. Land use/land cover (LULC) information and vegetation characteristics were extracted from satellite imagery and used to derive cover-management (C) and conservation-practice (P) factors. All spatial datasets (Table 1) were projected to the WGS 84/UTM Zone 43 N coordinate reference system and resampled to a common spatial resolution of 30 m prior to analysis. Raster processing, factor generation, overlay analysis, and erosion-risk mapping were performed using ArcGIS software. This ensured spatial consistency among all RUSLE input layers and minimized uncertainties arising from differences in spatial resolution and projection systems.

The integration of these datasets enabled a spatially explicit and cost-effective assessment of soil erosion risk at the sub-watershed scale. Despite limitations associated with generalized soil and land-use datasets, the selected inputs are widely adopted in watershed-scale RUSLE studies and provide a reliable basis for erosion-risk assessment.

2.3 Methodology

2.3.1 Soil erosion modelling approach

Soil erosion models are generally classified into process-based and empirical approaches. Process-based models simulate the physical mechanisms governing erosion, including rainfall impact, runoff generation, soil detachment, transport, and deposition. Although these models provide detailed process representation, they require extensive datasets and substantial computational resources, limiting their applicability in data-scarce regions [77]. In contrast, empirical models rely on statistical relationships derived from observed erosion data and are widely used because of their simplicity, robustness, and moderate data requirements. Several empirical and semi-empirical models have been developed, including WEPP, ANSWERS, LISEM, EUROSEM, SWAT, and SEMMED [78–80]. Among these, the Revised Universal Soil Loss Equation (RUSLE) remains one of the most widely applied models for estimating long-term average annual soil loss, particularly when integrated with Geographic Information Systems (GIS) [81].

2.3.2 Revised Universal Soil Loss Equation (RUSLE)

The Revised Universal Soil Loss Equation (RUSLE) is an empirical model designed to estimate long-term average annual soil loss resulting from rainfall-induced erosion under varying environmental and land-management conditions [82]. The model incorporates five primary factors: rainfall erosivity (R), soil erodibility (K), slope length and steepness (LS), cover management (C), and conservation practice (P). Each factor was represented as a raster layer within the GIS environment, allowing pixel-wise computation of soil erosion across the watershed.

The RUSLE model is expressed as:

Table 1 Summary of datasets used in the study

Dataset	Source	Resolution/Scale	Purpose
Rainfall (2014–2023)	IMD	Monthly	R-factor
Soil Map	Soil Survey of Kerala	1:50,000	K-factor
SRTM DEM	USGS/NASA	30 m	LS-factor
Satellite Imagery	Landsat/Sentinel	10–30 m	NDVI, C-factor
LULC Map	Classified imagery	10–30 m	P-factor

$$A = R \times K \times LS \times C \times P \tag{1}$$

where, A=estimated annual soil loss (t ha⁻¹ yr⁻¹). R=rainfall erosivity factor. K=soil erodibility factor. LS=slope length and steepness factor. C=cover management factor. P=conservation practice factor.

The adaptability and proven performance of RUSLE make it particularly suitable for regional and sub-watershed-scale erosion assessments in tropical environments [59–61].

2.3.3 Rainfall erosivity factor (R)

Rainfall erosivity (R) represents the capacity of rainfall to detach and transport soil particles and is influenced by both rainfall amount and intensity. In this study, the R-factor was calculated using ten years (2014–2023) of monthly rainfall data obtained from IMD. Due to the absence of detailed pluviographic records, the Modified Fournier Index (MFI) method was adopted [83].

$$R = \sum \left[1.735 \times 10^{(1.5 \times \log \left(\frac{P_i}{P} \right)^2 - 0.8188)} \right] \tag{2}$$

where: R=rainfall erosivity factor (MJ mm ha⁻¹ h⁻¹ yr⁻¹); P=total annual precipitation (mm); Pi= precipitation in month i (mm).

The resulting rainfall erosivity values were spatially interpolated to generate a continuous R-factor surface for the watershed. This approach has been widely applied in tropical and monsoon-dominated environments where rainfall intensity records are unavailable [84].

2.3.4 Soil erodibility factor (K)

The soil erodibility factor (K) represents the inherent susceptibility of soil to detachment and transport under rainfall and runoff conditions. Soil texture information was obtained from the Soil Survey of Kerala at a scale of 1:50,000. Although suitable for watershed-level analysis, the soil map represents generalized soil units and may not capture local-scale variability in soil properties. The assigned K-values were selected from published studies conducted in tropical and lateritic soil environments similar to those found in Kerala. The assigned values ranged from 0.019 to 0.0445 depending on soil texture class and were subsequently integrated within the GIS environment to generate a continuous spatial K-factor surface representing relative variations in soil erodibility across the watershed.

Although field-based measurements provide greater accuracy, the use of generalized soil maps and literature-derived K-values (Table 2) is a common and accepted practice in watershed-scale RUSLE studies, particularly in data-limited regions [73].

Table 2 Assigned soil erodibility (K) values based on soil texture classes

Soil Texture Class	Assigned K-value
Sandy and Sandy Loam	0.024
Loamy Sand	0.019
Sandy Soils	0.02
Clay	0.0296
Loam	0.0445

2.3.5 Slope length and steepness factor (LS)

The LS-factor accounts for the influence of topography on erosion by combining slope gradient and slope length [51]. Steeper and longer slopes increase runoff velocity and erosive power, thereby enhancing soil loss. The LS-factor was derived from the 30 m SRTM DEM using hydrologically corrected flow accumulation and slope layers. Computation followed the Desmet and Govers (1996) approach with $m = 0.5$ and $n = 1.3$.

$$LS = \frac{(\text{Flow Accumulation} \times \text{Cell Size})^m}{22.13} \times \frac{(\sin \text{Slope} \times 0.01745)^n}{0.0896} \quad (3)$$

where: Flow Accumulation = number of upstream contributing cells. Cell Size = DEM resolution (m). Slope = slope angle (degrees). m = slope length exponent (0.5). n = slope steepness exponent (1.3).

2.3.6 Cover management factor (C)

The cover management factor (C) reflects the protective influence of vegetation cover and land-management practices against soil erosion. The C-factor was derived from the Normalized Difference Vegetation Index (NDVI) calculated from satellite imagery. An exponential relationship between NDVI and C-factor was applied:

$$C = \exp \left(-2 \times \left(\frac{\text{NDVI}}{1 - \text{NDVI}} \right) \right)^2 \quad (4)$$

where higher NDVI values correspond to lower C-factor values and therefore lower erosion susceptibility. The derived C-values were consistent with values reported for forests, plantations, agricultural land, and fallow areas in tropical environments [84–89].

2.3.7 Conservation practice factor (P)

The conservation practice factor (P) represents the effectiveness of soil and water conservation measures in reducing erosion relative to conventional cultivation practices. P-factor values range between 0 and 1, with lower values indicating more effective erosion control [72]. P-values (Table 3) were assigned based on land-use classes and prevailing conservation practices within the watershed. Published RUSLE studies and commonly adopted conservation-practice coefficients were used to guide P-factor assignment. In areas where specific conservation structures were absent, literature-based values were adopted to represent existing management conditions. The resulting P-factor layer was

Table 3 Assigned conservation practice (P) values for land-use classes

LULC classes	P value
Barren rock	0.2
Built-up area	0
Deciduous forest	1
Land with scrub	0.9
Land without scrub	0.4
Plantation	0.5
River/Waterbody	0
Sandy area	0.3
Wetlands	0.1

Table 4 Comparison of soil loss estimates with similar studies

Study	Location	Method	Mean Soil Loss ($\text{t ha}^{-1} \text{yr}^{-1}$)
Jose et al.	Kerala	RUSLE-GIS	14.8
Thomas et al.	Western Ghats	RUSLE	19.6
Suresh et al.	Tropical India	RUSLE-GIS	16.3
Present Study	Peruvamba	RUSLE-GIS	17.4

Table 5 Sensitivity of soil loss to variations in RUSLE factors

Parameter	Variation	Change in Soil Loss (%)
R	$\pm 10\%$	± 9.8
K	$\pm 10\%$	± 10.1
LS	$\pm 10\%$	± 15.6
C	$\pm 10\%$	± 13.9

integrated within the GIS environment to identify areas where conservation interventions are either effective or insufficient.

2.3.8 Model validation and uncertainty assessment

Direct field-based measurements of soil loss were unavailable for validation. Therefore, model reliability was evaluated using sensitivity analysis and comparison with published erosion estimates from similar watersheds in Kerala and the Western Ghats (Table 4). Sensitivity analysis was performed by varying key RUSLE factors (R, K, LS, and C) by ± 10 – 20% and evaluating the resulting changes in estimated soil loss. The results indicated that LS and C factors exerted the greatest influence on model outputs, whereas K-factor variations produced comparatively moderate responses. These findings are consistent with previous studies in tropical mountainous environments where terrain and vegetation cover are dominant controls on erosion processes. A broader comparison with published RUSLE-based soil erosion studies from the Western Ghats is provided in Supplementary Table S1.

The estimated mean annual soil loss of $17.4 \text{ t ha}^{-1} \text{yr}^{-1}$ was compared with values reported for comparable Western Ghats watersheds (14.8 – $19.6 \text{ t ha}^{-1} \text{yr}^{-1}$), demonstrating close agreement. This comparison provides indirect validation of the model outputs and supports the reliability of the generated erosion-risk maps. The sensitivity analysis results are summarized in Table 5.

It should be noted that the RUSLE framework primarily represents sheet and rill erosion processes and does not explicitly account for gully erosion, channel erosion, mass wasting, or sediment deposition. Consequently, the predicted soil-loss values should be interpreted as indicators of relative erosion susceptibility rather than direct measurements of sediment yield. The overall methodological framework adopted for GIS-based RUSLE soil erosion assessment is presented in Fig. 2.

3 Results

The RUSLE model integrated within a GIS environment was used to estimate the spatial distribution of soil erosion risk in the Peruvamba sub-watershed. The generated factor maps and erosion-risk classifications revealed substantial spatial variability in erosion susceptibility driven by rainfall, soil characteristics, topography, vegetation cover, and land-management practices.

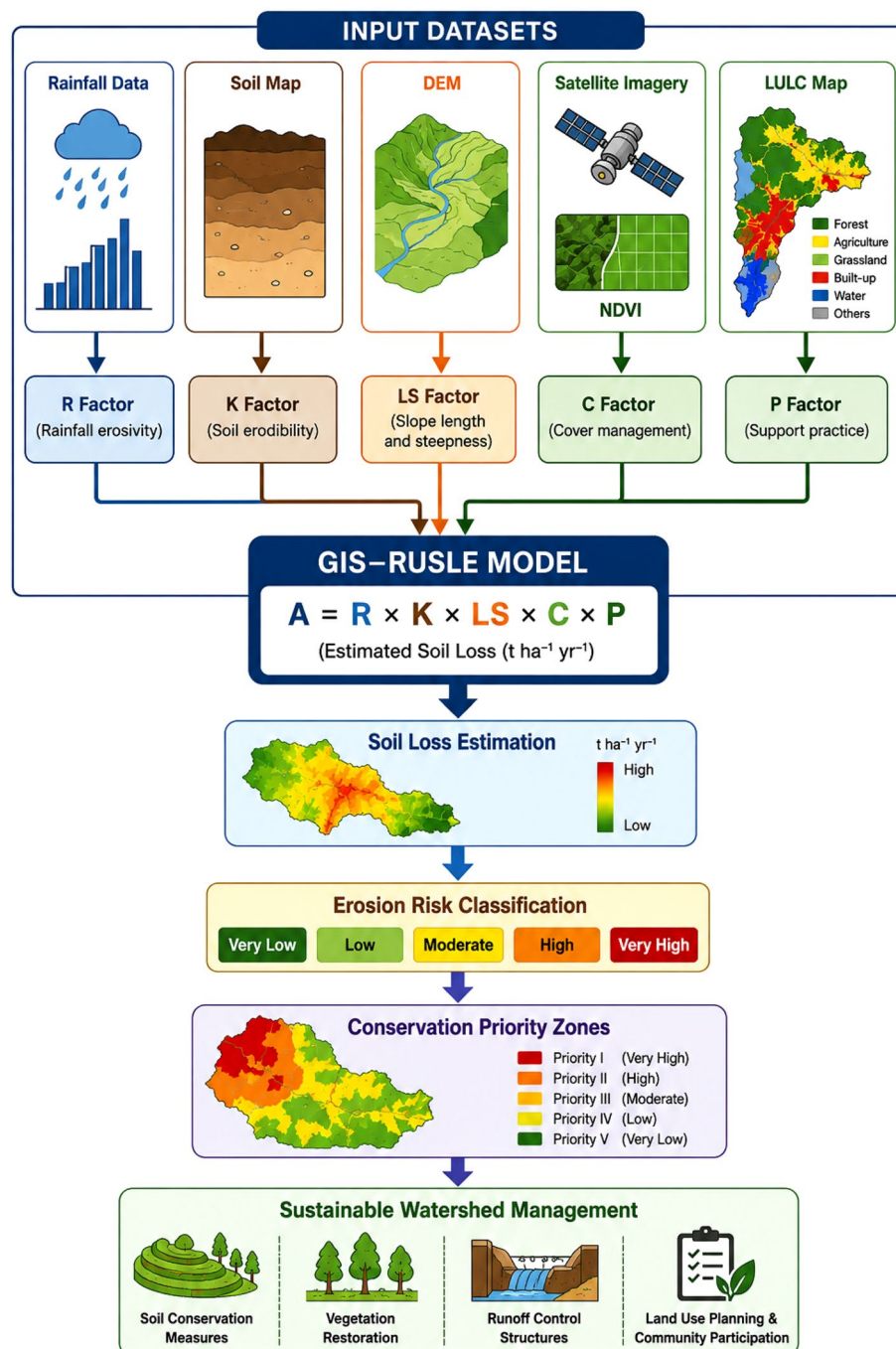


Fig. 2 Methodological workflow for GIS-based RUSLE soil erosion assessment. Multiple spatial datasets including rainfall, soil, topography, vegetation, and land-use information were integrated within a GIS environment to derive the R, K, LS, C, and P factors. These factors were combined using the RUSLE model to estimate annual soil loss, classify erosion risk, identify conservation priority zones, and support sustainable watershed management

3.1 Rainfall erosivity factor (R)

Rainfall erosivity (R) represents the erosive potential of rainfall and constitutes a major driver of soil detachment and transport. The computed R-factor values ranged from 1785.31 to 3812.84 $\text{MJ mm ha}^{-1} \text{h}^{-1} \text{yr}^{-1}$, with a mean value of 2710.67 $\text{MJ mm ha}^{-1} \text{h}^{-1} \text{yr}^{-1}$ across the study period (2014–2023) (Fig. 3). The observed variability reflects fluctuations in monsoonal rainfall intensity and distribution. The highest erosivity values

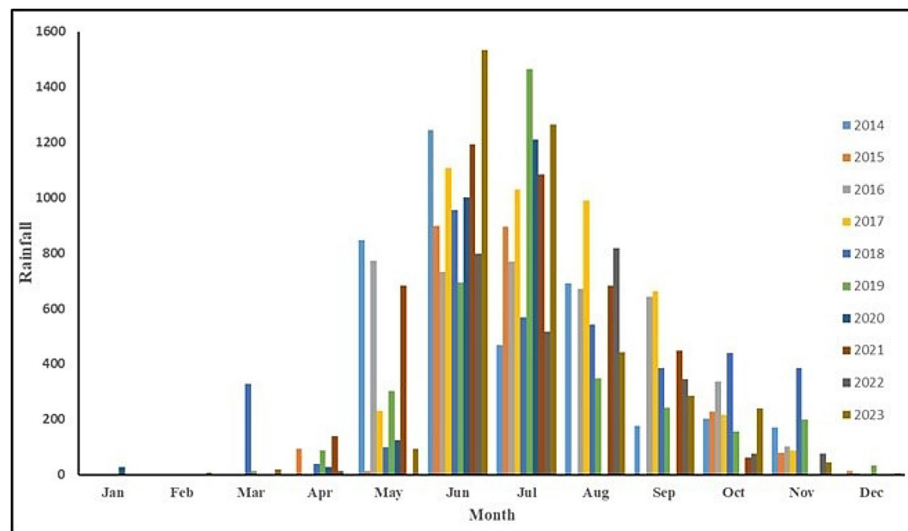


Fig. 3 Monthly rainfall data and annual average rainfall erosivity factor (R) for the period 2014–2023

corresponded to years with above-average monsoon rainfall, whereas lower values were associated with relatively dry years. Monthly rainfall analysis indicated that the majority of erosive rainfall occurred during the southwest monsoon season (June–September), suggesting that erosion risk is strongly seasonal. The magnitude of the estimated R-factor is comparable to values reported for other humid tropical watersheds of the Western Ghats, indicating a substantial potential for rainfall-induced erosion within the watershed.

3.2 Soil erodibility factor (K)

The soil erodibility factor (K) reflects the inherent susceptibility of soils to erosion under rainfall and runoff conditions. Based on soil texture information, assigned K-values ranged from 0.019 to 0.0445, indicating moderate spatial variability in soil erodibility across the watershed (Fig. 4). Higher K-values were associated with loamy and sandy-loam soils that are more susceptible to particle detachment, whereas lower values occurred in clay-rich soils characterized by greater structural stability. Areas exhibiting relatively high K-values were primarily distributed in agricultural and partially degraded landscapes where soil properties increase vulnerability to erosion [91]. Although generalized soil maps were used, the derived K-values are consistent with those reported for tropical lateritic soils in southern India and provide a reasonable representation of soil erodibility at the watershed scale.

3.3 Slope length and steepness factor (LS)

The slope length and steepness factor (LS) represents the influence of topography on runoff generation and erosion processes. LS-values ranged from 0.0 to 28.88, reflecting pronounced topographic heterogeneity within the watershed (Fig. 5). High LS-values were concentrated in the upland and hilly regions, particularly in the southern, western, and central portions of the watershed [93]. These areas are characterized by steep slopes and long flow paths that promote rapid runoff concentration and increase the erosive power of surface flow. In contrast, valley bottoms and gently sloping terrain exhibited low LS-values and correspondingly lower erosion susceptibility [94]. The spatial

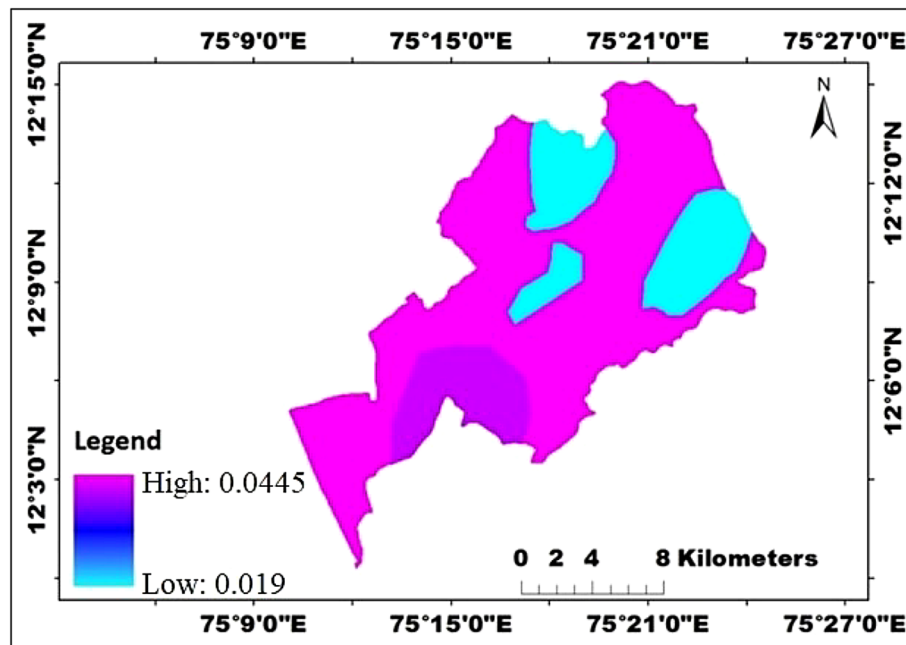


Fig. 4 Spatial distribution of soil erodibility factor (K)

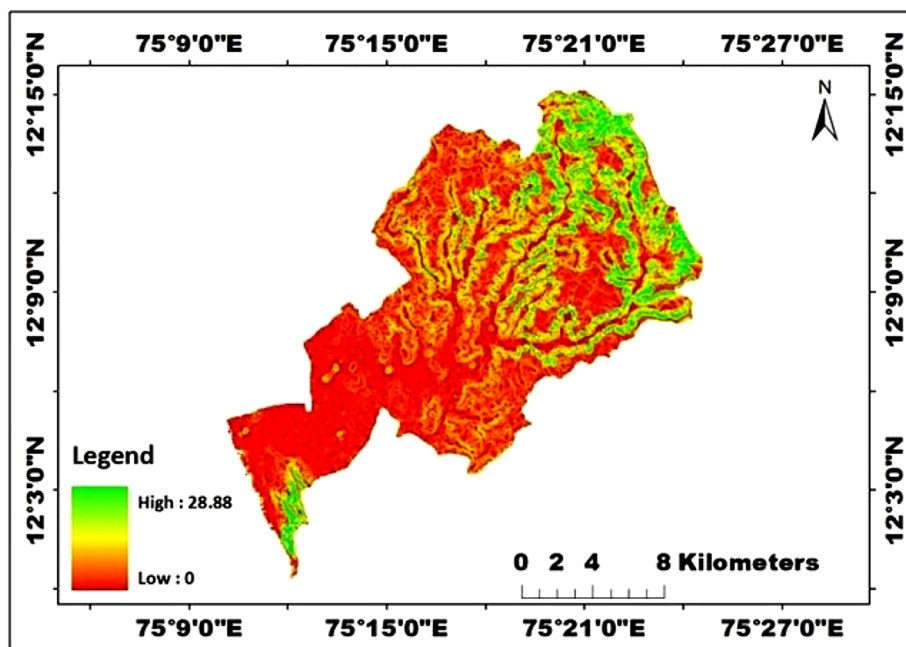


Fig. 5 Spatial distribution of slope length and steepness factor (LS)

correspondence between elevated LS-values and areas of high soil loss highlights topography as a dominant control on erosion processes in the watershed.

3.4 Cover management factor (C)

The cover management factor (C) quantifies the protective influence of vegetation cover against soil erosion. C-factor values derived from NDVI ranged from 0.002 to 0.25 (Fig. 6). Very low C-values were associated with dense forests and plantation areas,

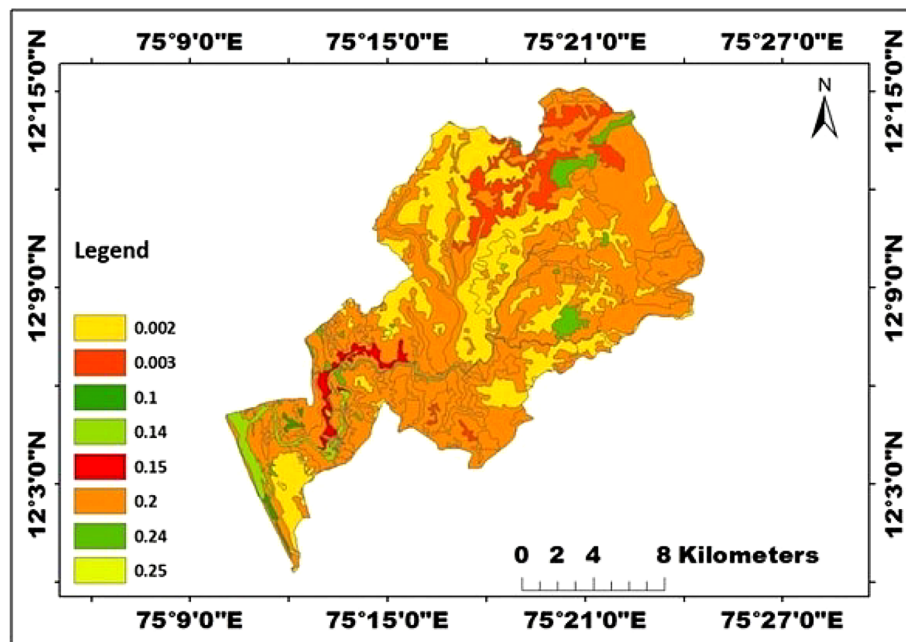


Fig. 6 Spatial distribution of cover management factor (C)

where canopy cover, litter accumulation, and root systems effectively reduce raindrop impact and surface runoff. Moderate values occurred in agricultural lands with seasonal vegetation cover, while high C-values were observed in degraded lands, scrublands, and sparsely vegetated areas. The spatial distribution of C-values clearly demonstrates the role of vegetation cover in mitigating soil erosion. Areas with dense vegetation consistently exhibited lower erosion susceptibility, whereas degraded and exposed landscapes showed elevated erosion risk.

3.5 Conservation practice factor (P)

The conservation practice factor (P) represents the effectiveness of land-management interventions in reducing soil erosion. Assigned P-values varied between 0.0 and 1.0 according to land-use characteristics and prevailing management practices (Table 3). Low P-values were assigned to built-up areas and water bodies where sheet and rill erosion are negligible. Intermediate values characterized plantations, wetlands, sandy areas, and lands without scrub, while higher values were associated with scrublands and deciduous forests where specific erosion-control measures were absent. The resulting P-factor distribution identifies areas where improved conservation practices could substantially reduce erosion risk [95–96].

3.6 Land use/land cover characteristics

The land use/land cover (LULC) map revealed a heterogeneous landscape comprising forests, plantations, scrublands, agricultural lands, wetlands, built-up areas, sandy zones, barren rock surfaces, and river channels (Fig. 7). Forests and plantations dominated the central and southern portions of the watershed, while scrublands and degraded lands were more common in the northern and western sectors [97–98]. Overlay analysis indicated a strong relationship between land-cover characteristics and erosion patterns. Areas with sparse vegetation cover frequently coincided with steep slopes and elevated

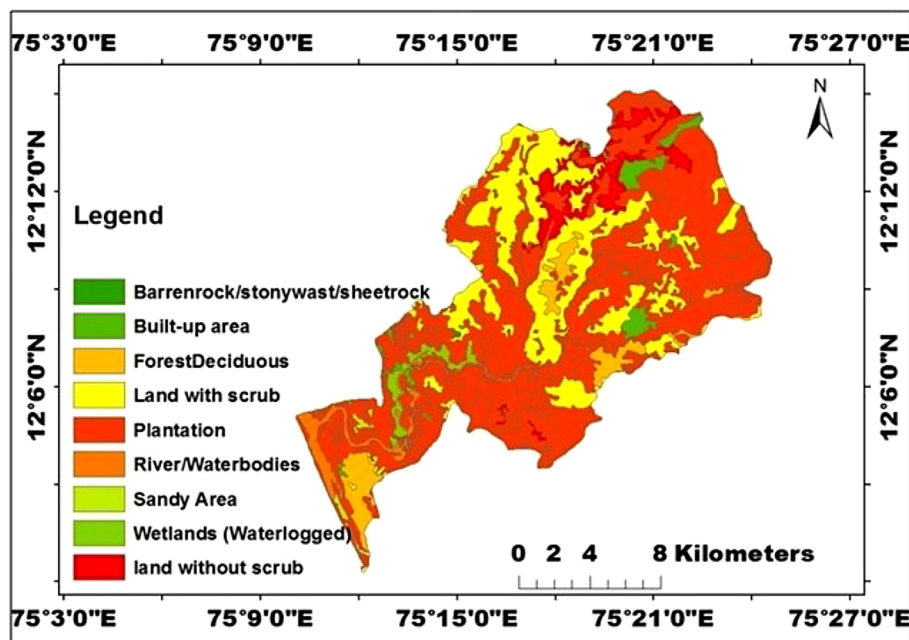


Fig. 7 Land use/land cover map of the Peruvamba sub-watershed

LS-values, resulting in increased erosion susceptibility. In contrast, forested and plantation areas exhibited lower erosion rates due to effective soil protection and root reinforcement. These observations demonstrate the significant influence of land-use patterns on erosion processes within the watershed.

3.7 Soil erosion risk assessment

Integration of all RUSLE factors produced a spatially explicit soil erosion risk map for the Peruvamba sub-watershed (Fig. 8). Estimated annual soil loss values were classified into five erosion-severity classes: very low ($< 5 \text{ t ha}^{-1} \text{ yr}^{-1}$), low ($5\text{--}10 \text{ t ha}^{-1} \text{ yr}^{-1}$), moderate ($10\text{--}20 \text{ t ha}^{-1} \text{ yr}^{-1}$), high ($20\text{--}40 \text{ t ha}^{-1} \text{ yr}^{-1}$), and very high ($> 40 \text{ t ha}^{-1} \text{ yr}^{-1}$). The total estimated annual soil loss was approximately $72,500 \text{ t yr}^{-1}$, with a mean annual soil loss of $17.4 \text{ t ha}^{-1} \text{ yr}^{-1}$. This value exceeds the commonly accepted soil-loss tolerance threshold of $10\text{--}12 \text{ t ha}^{-1} \text{ yr}^{-1}$ for tropical regions, indicating the need for targeted conservation interventions [99–100]. The spatial distribution of erosion severity classes is presented in Table 6. Approximately 41.2% (110.83 km^2) of the watershed fell within the very low erosion class, followed by 26.3% (70.75 km^2) under low risk, 17.8% (47.88 km^2) under moderate risk, 9.4% (25.29 km^2) under high risk, and 5.3% (14.26 km^2) under very high risk. High and very high erosion classes (Table 6) together occupied 39.55 km^2 (14.7%) of the watershed and were concentrated primarily in steep upland regions characterized by sparse vegetation cover and relatively high soil erodibility. These areas represent priority zones for soil and water conservation measures.

3.8 Seasonal Rainfall characteristics and erosion implications

Monthly rainfall distribution exhibited a strong seasonal concentration associated with the southwest monsoon (Fig. 9). Peak rainfall occurred during June and July, accounting for a substantial proportion of annual precipitation. Although monthly soil loss was not explicitly calculated, the rainfall pattern indicates that erosion risk is likely to be highest

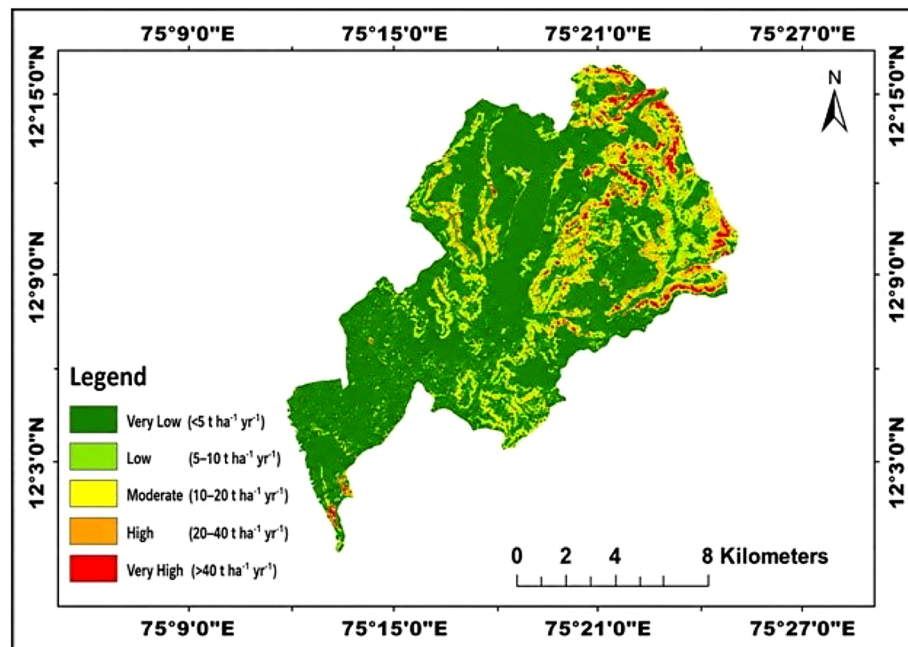


Fig. 8 Spatial distribution of annual soil erosion risk classes in the Peruvamba sub-watershed

Table 6 Soil erosion severity classes and area distribution

Class	Soil Loss ($\text{t ha}^{-1} \text{yr}^{-1}$)	Area (km^2)	Area (%)
Very Low	< 5	110.83	41.2
Low	5–10	70.75	26.3
Moderate	10–20	47.88	17.8
High	20–40	25.29	9.4
Very High	> 40	14.26	5.3
Total	—	269	100

during the monsoon period when runoff generation and sediment transport are most intense. This seasonal concentration highlights the importance of implementing erosion-control measures prior to the onset of the monsoon season.

3.9 Validation and sensitivity analysis

The reliability of the RUSLE outputs was evaluated through comparison with erosion estimates reported for similar tropical watersheds and through sensitivity analysis of key model parameters. The estimated mean annual soil loss of $17.4 \text{ t ha}^{-1} \text{yr}^{-1}$ falls within the range reported for comparable Western Ghats watersheds ($14.8\text{--}19.6 \text{ t ha}^{-1} \text{yr}^{-1}$), indicating good agreement with previous studies. Sensitivity analysis revealed that the relative influence of RUSLE factors on soil-loss variability followed the order:

$$\text{LS} > \text{C} > \text{K} > \text{R}$$

This ranking indicates that topography and vegetation cover exert the strongest influence on erosion processes within the watershed. Consequently, management interventions focused on slope stabilization and vegetation restoration are likely to achieve the greatest reduction in soil-loss rates. The agreement between predicted erosion rates, spatial patterns of RUSLE factors, and published observations from comparable tropical

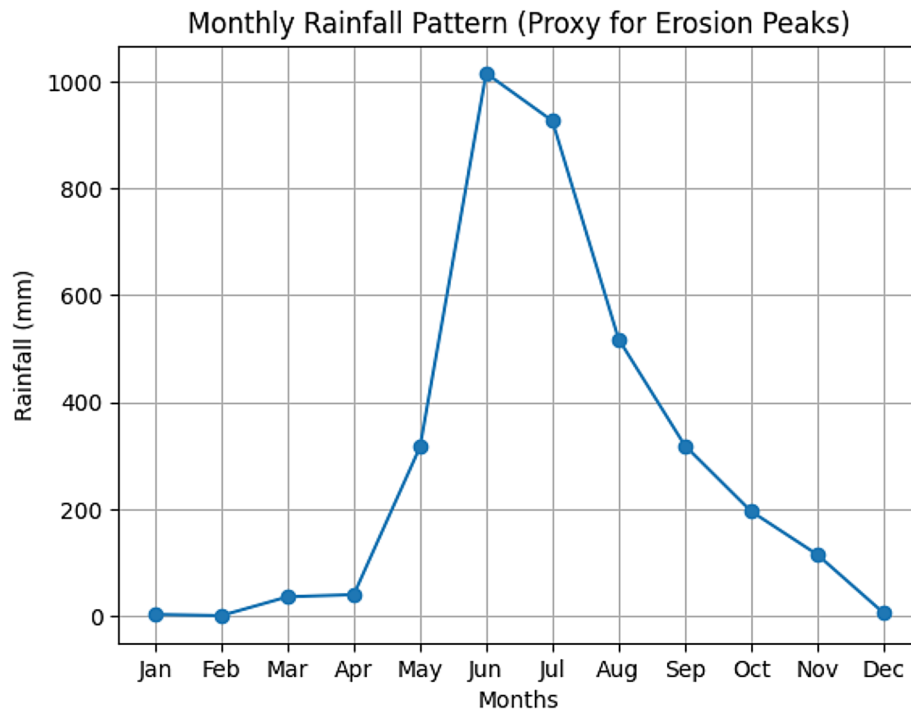


Fig. 9 Monthly rainfall distribution indicating seasonal concentration of rainfall and associated erosion potential

environments supports the reliability of the generated erosion-risk maps and confirms the suitability of the GIS-integrated RUSLE framework for watershed-scale erosion assessment. Although direct field measurements of soil loss were unavailable, the consistency between estimated erosion rates, observed topographic controls, and published studies from similar Western Ghats environments provides confidence in the spatial reliability of the generated erosion-risk maps. The sensitivity analysis demonstrated that a $\pm 10\%$ variation in RUSLE input factors resulted in soil-loss changes ranging from $\pm 9.8\%$ to $\pm 15.6\%$, indicating moderate model sensitivity. Among the evaluated factors, topographic characteristics (LS) exhibited the highest influence on model outputs, whereas rainfall erosivity (R) showed comparatively lower sensitivity. These findings suggest that the spatial patterns of erosion risk identified in this study are robust despite uncertainties associated with input datasets.

4 Discussion

The spatial assessment of soil erosion in the Peruvamba sub-watershed revealed that erosion dynamics are controlled by the combined influence of rainfall erosivity, topography, soil properties, vegetation cover, and conservation practices. The integration of RUSLE with GIS and remote sensing enabled the identification of erosion hotspots and provided a detailed understanding of the spatial variability of erosion risk within the watershed. Areas classified as high and very high erosion risk were primarily associated with steep slopes, elevated rainfall erosivity, sparse vegetation cover, and inadequate conservation practices. In contrast, forested and wetland areas exhibited comparatively low soil-loss rates owing to dense vegetation cover, improved soil stability, and reduced runoff generation. The strong correspondence between erosion-risk patterns and the

controlling environmental factors demonstrates the effectiveness of the GIS-integrated RUSLE framework for erosion assessment in humid tropical watersheds.

4.1 Influence of monsoonal rainfall on erosion dynamics

Rainfall erosivity emerged as a major driver of soil erosion in the Peruvamba sub-watershed. The calculated R-factor values indicate that the watershed experiences substantial erosive energy associated with monsoonal rainfall. The concentration of rainfall during the southwest monsoon season (June–September) significantly increases runoff generation and enhances the detachment and transport of soil particles. Similar relationships between monsoonal rainfall and erosion intensity have been reported in tropical watersheds across the Western Ghats and other monsoon-dominated regions. The observed inter-annual variability in rainfall erosivity suggests that erosion risk within the watershed is highly sensitive to fluctuations in rainfall intensity and distribution. This finding is particularly important under changing climatic conditions. Recent studies in the Western Ghats indicate increasing frequencies of extreme rainfall events, which can intensify runoff generation, accelerate sediment transport, and increase erosion rates. Consequently, future watershed management strategies should integrate climate-resilient soil conservation practices and focus on strengthening erosion-control measures before the onset of the monsoon season. The adoption of contour farming, vegetative barriers, check dams, and runoff-harvesting structures can substantially reduce erosion during periods of intense rainfall.

4.2 Role of topography in controlling soil loss

Topography exerted a strong influence on the spatial distribution of soil erosion across the watershed. The LS-factor showed substantial variability, with the highest values occurring in steep upland regions where runoff velocity and erosive energy are considerably enhanced. The concentration of high erosion-risk zones in areas with elevated LS-values confirms the dominant role of terrain in controlling soil-loss processes. The relationship between slope steepness and erosion is well documented in mountainous environments, where increased runoff velocity promotes greater soil detachment and sediment transport. In the Peruvamba sub-watershed, high LS-values coincided closely with areas classified under high and very high erosion risk, demonstrating the importance of topographic controls on erosion processes. These findings suggest that slope-specific conservation measures should be prioritized in upland regions. Structural interventions such as terracing, contour bunding, stone bunds, vegetative reinforcement, and check dams can significantly reduce runoff velocity and improve slope stability.

Based on the erosion-risk classification, the watershed can be divided into three management-priority categories. High and very high erosion-risk areas require immediate structural interventions and intensive conservation measures. Moderate-risk areas should be managed through preventive land-management practices such as agroforestry, contour cultivation, and cover cropping. Low-risk zones should focus on maintaining existing vegetation cover and monitoring future land-use changes to prevent degradation.

4.3 Vegetation cover and land use as protective factors

Vegetation cover was identified as one of the most important factors controlling soil erosion within the watershed. Areas characterized by dense forest cover and perennial plantations exhibited low C-factor values and correspondingly low soil-loss rates. Vegetation reduces the kinetic energy of raindrops, enhances infiltration, stabilizes soil through root reinforcement, and decreases surface runoff, thereby providing effective protection against erosion. Conversely, agricultural lands, scrublands, and degraded landscapes exhibited higher C-factor values and greater erosion susceptibility. Seasonal agricultural practices, land clearing, and reduced vegetation cover expose the soil surface to direct rainfall impact, increasing the likelihood of soil detachment and transport. These findings are consistent with previous studies conducted in tropical watersheds, which have demonstrated that conversion of forested land to agriculture or degraded scrubland substantially increases erosion risk. Analysis of mean soil loss across major land-use classes further highlighted the influence of vegetation cover on erosion dynamics (Fig. 10). Forested areas exhibited the lowest mean annual soil loss (approximately 3 t ha⁻¹ yr⁻¹), followed by plantations (approximately 8 t ha⁻¹ yr⁻¹). Agricultural lands experienced moderate erosion rates (approximately 16 t ha⁻¹ yr⁻¹), whereas scrublands and barren lands recorded substantially higher soil-loss values of approximately 24 and 38 t ha⁻¹ yr⁻¹, respectively. These findings emphasize the importance of maintaining continuous vegetation cover through afforestation, agroforestry, cover cropping, and sustainable land-management practices to reduce soil erosion and maintain watershed health.

4.4 Soil properties and management implications

Although the spatial variability of the soil erodibility factor (K) was lower than that of rainfall or topography, soil characteristics still played an important role in determining erosion susceptibility. Soils with higher sand and silt contents and lower organic matter levels were generally more vulnerable to detachment and transport. The interaction between soil erodibility, slope steepness, and vegetation cover amplified erosion risk in several parts of the watershed. Improving soil quality can therefore contribute significantly to erosion control. Practices such as organic amendments, residue retention, conservation tillage, mulching, and cover cropping can enhance soil structure, improve aggregate stability, increase infiltration, and reduce runoff generation. These measures complement structural conservation practices and provide long-term benefits for soil health and agricultural productivity.

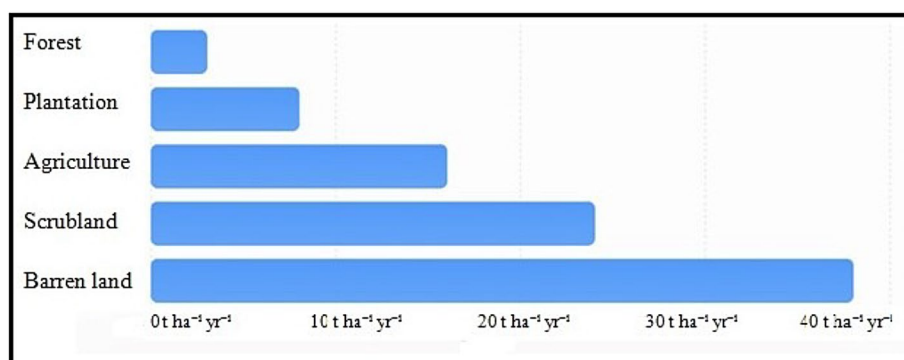


Fig. 10 Mean annual soil loss across major land use/land cover classes in the Peruvamba sub-watershed

4.5 Effectiveness of conservation practices and management gaps

The conservation practice factor (P; Table 3) revealed substantial variation in the effectiveness of existing land-management practices throughout the watershed. Areas characterized by low P-values generally corresponded to landscapes where erosion-control measures or natural protection mechanisms were effective. In contrast, scrublands and degraded areas with high P-values indicated limited conservation intervention and elevated erosion susceptibility. The spatial distribution of P-values suggests that conservation measures are not uniformly implemented across the watershed. Consequently, targeted conservation planning is required to maximize the effectiveness of erosion-control investments. Areas exhibiting high LS, high C, and high P values should be prioritized for immediate intervention through contour trenches, vegetative barriers, check dams, grassed waterways, and runoff-control structures. Such targeted measures are likely to achieve greater environmental benefits and cost-effectiveness than broad, uniform interventions.

4.6 Implications for watershed management and policy

The erosion-risk maps developed in this study provide a valuable decision-support framework for watershed management, conservation prioritization, and sustainable land-use planning. By identifying erosion hotspots and quantifying their severity, the study enables resource managers and policymakers to allocate conservation investments more effectively. Recommended conservation measures corresponding to different erosion severity classes are summarized in Table 7.

The proposed management framework emphasizes a risk-based approach in which conservation measures are matched to erosion severity. Such an approach can improve the efficiency of watershed restoration programmes while minimizing unnecessary expenditure in low-risk areas. In ecologically sensitive landscapes such as the Western Ghats, integrating erosion-risk assessments into regional development planning is essential for balancing economic development with environmental sustainability. The findings have broader relevance for watershed management across the Western Ghats, where steep terrain, highly weathered lateritic soils, and increasing land-use pressures contribute to widespread erosion problems. The methodological framework developed in this study can support conservation prioritization in similar tropical watersheds experiencing rapid environmental change and increasing hydro-climatic variability. Projected increases in rainfall intensity and extreme precipitation events under climate-change scenarios may further intensify erosion processes in the Western Ghats. Consequently, erosion-risk assessments should be periodically updated and integrated into climate-resilient watershed management strategies to support long-term ecosystem sustainability and agricultural productivity. The integrated watershed management framework developed from the present study is illustrated in Figure 11.

Table 7 Recommended conservation measures according to erosion severity

Erosion Class	Area (%)	Recommended Conservation Measure
Very Low	41.2	Monitoring and maintenance
Low	26.3	Vegetative cover enhancement
Moderate	17.8	Contour farming and mulching
High	9.4	Terracing and contour bunding
Very High	5.3	Check dams, slope stabilization, and immediate intervention

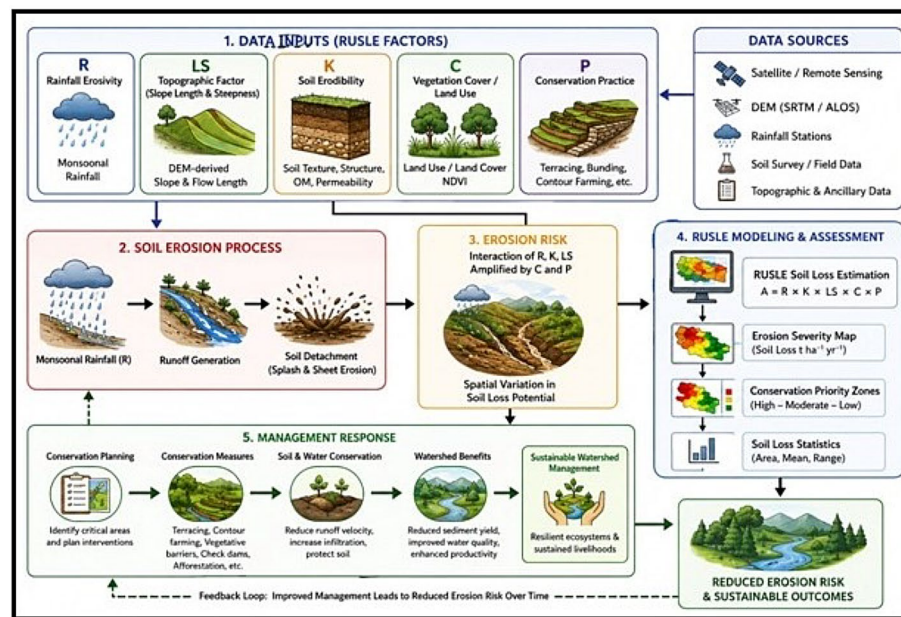


Fig. 11 Conceptual framework illustrating the interaction of rainfall erosivity (R), soil erodibility (K), topographic factors (LS), vegetation cover (C), and conservation practices (P) in controlling soil erosion processes and guiding sustainable watershed management. Projected increases in rainfall intensity and extreme precipitation events under future climate scenarios may further intensify erosion processes in the Western Ghats, emphasizing the need for climate-resilient watershed management strategies

4.7 Limitations and future research directions

Although the GIS-integrated RUSLE framework provides a practical and widely accepted approach for watershed-scale erosion assessment, several limitations should be acknowledged. The analysis relied on generalized soil maps, satellite-derived vegetation indices, and rainfall datasets lacking sub-daily intensity information. These limitations may introduce uncertainty in the estimation of RUSLE factors, particularly rainfall erosivity and soil erodibility. Furthermore, RUSLE primarily represents sheet and rill erosion processes and does not explicitly account for gully erosion, channel erosion, landslides, or sediment deposition processes, which may contribute significantly to sediment dynamics in steep tropical landscapes. Consequently, the predicted soil-loss values should be interpreted as indicators of relative erosion susceptibility rather than direct measurements of sediment yield. Despite these limitations, the estimated mean annual soil loss of $17.4 \text{ t ha}^{-1} \text{ yr}^{-1}$ falls within the range reported for comparable Western Ghats watersheds ($14.8\text{--}19.6 \text{ t ha}^{-1} \text{ yr}^{-1}$), providing indirect validation of the model outputs and supporting the reliability of the generated erosion-risk maps. Future studies should incorporate field-based sediment measurements, higher-resolution rainfall intensity datasets, and process-based hydrological models such as SWAT or WEPP to improve predictive accuracy. The integration of climate-change projections and future land-use scenarios would further strengthen erosion-risk assessments and support long-term watershed management planning. The recommended conservation measures for different erosion-risk categories are summarized in Table 8.

Table 8 Conservation priority matrix

Erosion Risk	Area (%)	Priority
Very High	5.3	Immediate
High	9.4	Short-term
Moderate	17.8	Medium-term
Low	26.3	Monitoring
Very Low	41.2	Maintenance

5 Conclusions

This study evaluated soil erosion risk in the Peruvamba sub-watershed of Kerala, India, through the integration of the Revised Universal Soil Loss Equation (RUSLE), Geographic Information Systems (GIS), and remote sensing techniques. The results demonstrated the capability of the RUSLE–GIS framework to effectively quantify the spatial variability of soil erosion and identify critical erosion-prone areas within a humid tropical watershed of the Western Ghats. The estimated total annual soil loss for the watershed was approximately 72,500 t yr⁻¹, with a mean annual soil loss of 17.4 t ha⁻¹ yr⁻¹, exceeding the commonly accepted soil-loss tolerance threshold for tropical regions. Approximately 41.2% of the watershed was classified as very low erosion risk, whereas 14.7% fell within high to very high erosion categories. These high-risk zones were primarily concentrated in steep upland areas characterized by sparse vegetation cover, high runoff potential, and limited conservation practices. In particular, approximately 5.3% of the watershed was identified as a very high erosion-risk zone requiring immediate conservation intervention. The analysis revealed that rainfall erosivity, topographic characteristics, vegetation cover, and conservation practices collectively control erosion dynamics within the watershed. Sensitivity analysis indicated that the relative influence of RUSLE factors followed the order $LS > C > K > R$, demonstrating that slope length–steepness and vegetation cover are the dominant controls on soil-loss variability. Forested and plantation areas exhibited comparatively low erosion rates, whereas scrublands, degraded lands, and exposed surfaces showed substantially greater susceptibility to erosion. These findings highlight the critical importance of maintaining vegetation cover and implementing slope-stabilization measures to reduce soil loss.

The generated erosion-risk maps provide a valuable decision-support tool for watershed management and conservation prioritization. Targeted interventions such as terracing, contour bunding, vegetative barriers, agroforestry, cover cropping, and check-dam construction should be prioritized in high-risk areas, particularly before the onset of the southwest monsoon season when erosion potential is greatest. Such measures can substantially reduce sediment loss, improve soil conservation, and enhance watershed resilience. Although the study relied primarily on secondary datasets and spatial modelling approaches, the estimated erosion rates were consistent with values reported for comparable Western Ghats watersheds, supporting the reliability of the results. Future studies should incorporate field-based validation, higher-resolution rainfall intensity data, and process-based hydrological models to improve predictive accuracy. The integration of climate-change scenarios and future land-use projections would further strengthen erosion-risk assessments and support the development of climate-resilient watershed management strategies. The study demonstrates that the integration of RUSLE, GIS, and remote sensing provides a robust, scalable, and transferable framework for soil erosion assessment in data-scarce tropical environments. The methodology and

findings can support evidence-based land-use planning, sustainable watershed management, and long-term soil conservation efforts across the Western Ghats and other erosion-prone regions.

Supplementary Information

The online version contains supplementary material available at <https://doi.org/10.1007/s44288-026-00622-2>.

Supplementary Material 1

Acknowledgements

The authors sincerely acknowledge Kannur University, Siddhartha Academy of Higher Education (Deemed to be University), Andhra Pradesh, and PSG College of Arts and Science for their support and facilities provided during the conduct of this research.

Author contributions

Sapna Kinattinkara- contributed to methodology design and evaluated the performance of the implemented project system, Thangavelu Arumugam - identified and specified the theoretical framework and methods, and wrote and reviewed the manuscript, Vinaykumar Gaddam - supported the methodology, assisted in the manuscript review process, and provided additional information, Karpagamani Vellingiri- implemented, formulated the designed work system for the study, Manimaran Arumugam - theoretically rewritten, conceptualization, and methods inputs.

Funding

The authors received no specific funding for this research.

Data availability

The datasets generated and/or analysed during the current study are available from Dr. Sapna (sapnakinattinkara@gmail.com) and Dr. Thangavelu Arumugam (thangavelgis@gmail.com), upon reasonable request. Publicly available datasets used in this study include the Shuttle Radar Topography Mission (SRTM) Digital Elevation Model (DEM) obtained from the United States Geological Survey (USGS). Rainfall data were obtained from the India Meteorological Department (IMD) and are subject to the applicable data-sharing policies of the IMD.

Declarations

Ethics approval and consent to participate

Not applicable. This study did not involve human participants, human tissues, animals, or any procedures requiring ethical approval. This study did not involve human participants or human data.

Consent for publication

Not applicable. The manuscript does not contain any identifiable personal information, images, or data relating to individual participants.

Competing interests

The authors declare no competing interests.

Received: 20 January 2026 / Accepted: 1 July 2026

Published online: 20 July 2026

References

1. Alsafadi K, Bi S, Abdo HG, Al Sayah MJ, Ratonyi T, Harsanyi E, et al. Spatial-temporal dynamic impact of changes in rainfall erosivity and vegetation coverage on soil erosion in the Eastern Mediterranean. *Environ Sci Pollut Res Int*. 2022. <https://doi.org/10.1007/s11356-022-24012-6>.
2. Olika G, Fikadu G, Gedefa B. GIS based soil loss assessment using RUSLE model: A case of Horo district, western Ethiopia. *Heliyon*. 2023;9(2):e13313. <https://doi.org/10.1016/j.heliyon.2023.e13313>.
3. Ghosh A, Rakshit S, Tikle S, Das S, Chatterjee U, Pande CB, Alataway A, Al-Othman AA, Dewidar AZ, Mattar MA. Integration of GIS and Remote Sensing with RUSLE Model for Estimation of Soil Erosion. *Land*. 2023;12(1):116. <https://doi.org/10.3390/land12010116>.
4. Shekhar PR, Mathew A. Gis-based assessment of soil erosion and sediment yield using the revised universal soil loss equation (RUSLE) model in the murredu watershed. *Telangana India HydroResearch*. 2024;7:315–25. <https://doi.org/10.1016/j.hydres.2024.05.003>.
5. Thangavelu Arumugam S, Kinattinkara S, Velusamy M, Shanmugamoorthy S Veerasamy. Assessment of hydro-logical behavioural changes of Noyyal watershed in Coimbatore district, India by using SWAT model. *Urban Clim*. 2025a;59:102285. <https://doi.org/10.1016/j.uclim.2025.102285>.
6. Jothamani M, Getahun E, Abebe A. Remote sensing, GIS, and RUSLE in soil loss estimation in the Kulfo river catchment, Rift valley, Southern Ethiopia Remote sensing, GIS, and RUSLE in soil loss estimation in the Kulfo river catchment, Rift valley. *South Ethiopia*. 2022. <https://doi.org/10.15243/jdmlm.2022.092.3307>.
7. Moisa MB. Soil loss estimation and prioritization using geographic information systems and the RUSLE model: a case study of the Anger River sub-basin Western Ethiopia. *J Water Clim Change*. 2022;13(3):1170–84. <https://doi.org/10.2166/wcc.2022.433>.

8. Ghosh J, Das S, Ghosh S, Midya K, Das S, Mishra VN. Soil erosion vulnerability analysis of Damodar River Basin, India using Revised Universal Soil Loss Equation (RUSLE) in Google Earth Engine (GEE). *Discover Geoscience*. 2025;3:1.
9. Tang Q, Xu Y, Bennett SJ, Li Y. Assessment of soil erosion using RUSLE and GIS: a case study of the Yangou watershed in the Loess Plateau, China. *Environ Earth Sci*. 2015;73(4):1715–24.
10. Uddin K, Abdul Matin M, Maharjan S. Assessment of land cover change and its impact on changes in soil erosion risk in Nepal. *Sustainability*. 2018;10(12):4715.
11. Thangavelu Arumugam SK, Dewali, Snehmani S Kinattinkara. Estimation of snow depth in GIS environment from observation points on Z Gali region: A case study of NW Himalaya. *Results Eng*. 2024. <https://doi.org/10.1016/j.rineng.2024.101869>. 101869, ISSN 2590–1230.
12. Teshome DS, Moisa MB, Gemedo DO, You S. Effect of land use-land cover change on soil erosion and sediment yield in Muger Sub-Basin, Upper Blue Nile Basin, Ethiopia. *Land*. 2022;11:2173.
13. Kumar A, Singh S, Patley MK, Memon F, Singh RK, Kinattinkara S, Thangavelu Arumugam. Integrated GIS-Based Aquifer Management System: A case study of Rajnandgaon District, Chhattisgarh, India. *J Hazard Mater Adv*. 2024;100586:2772–4166. <https://doi.org/10.1016/j.hazadv.2024.100586>.
14. Bastola S, Seong YJ, Lee SH, Shin Y. Assessment of soil erosion loss by using RUSLE and GIS in the Bagmati Basin of Nepal. *J Korean GEO-Environ Soc*. 2019;20(3):5–14.
15. Thapa P. Spatial estimation of soil erosion using RUSLE modeling: a case study of Dolakha district, Nepal. *Environ Syst Res*. 2020;9:15. <https://doi.org/10.1186/s40068-020-00177-2>.
16. Markose VJ, Jayappa KS. Soil loss estimation and prioritization of sub-watersheds of Kali River basin, Karnataka, India, using RUSLE and GIS. *Environ Monit Assess*. 2016;188:225.
17. Balabathina VN, Raju RP, Mulualem W, Tadele G. Estimation of soil loss using remote sensing and GIS-based universal soil loss equation in northern catchment of Lake Tana Sub-basin, Upper Blue Nile Basin, Northwest Ethiopia. *Environ Syst Res*. 2020. <https://doi.org/10.1186/s40068-020-00203-3>.
18. Khan A, Govil H. Evaluation of potential sites for soil erosion risk in and around Yamuna River flood plain using RUSLE. *Arab J Geosci*. 2020;13:707.
19. Sapna K, Thangavelu A, Kaarmuhil SP. Surveillance of groundwater quality of selected rural and industrial areas of Coimbatore: a GIS Approach. *IOP Conf Ser Mater Sci Eng*. 2020;762:955. <https://doi.org/10.1088/1757-899X/955/1/012083>.
20. Zhao J, Feng X, Deng L, Yang Y, Zhao Z, Zhao P, Peng C, Fu B. Quantifying the effects of vegetation restorations on the soil erosion export and nutrient loss on the Loess Plateau. *Front Plant Sci*. 2020;11:1–13. <https://doi.org/10.3389/fpls.2020.573126>.
21. Ghosh A, Maiti R. Soil erosion susceptibility assessment using logistic regression, decision tree and random forest: Study on the Mayurakshi river basin of Eastern India. *Environ Earth Sci*. 2021;80:328.
22. Thangavelu A, Manoj K, Sapna K, Jyothin CK, Prashanth KP. Investigation of land use cover patterns of sea shore vegetation of Kannur Coast of Northern Kerala, India using GIS. *Ecol Environ Conserv EM Int*. 2021a;27:S225–35.
23. Duguma TA. Soil erosion risk assessment and treatment priority classification: a case study on Guder watersheds, Abay River basin, Oromia, Ethiopia. *Heliyon*. 2022;8(8):e10183. <https://doi.org/10.1016/j.heliyon.2022.e10183>.
24. Yeneneh N, Elias E, Feyisa GL. Quantify soil erosion and sediment export in response to land use/cover change in the Suha watershed, northwestern highlands of Ethiopia: implications for watershed management. *Environ Syst Res*. 2022. <https://doi.org/10.1186/s40068-022-00265-5>.
25. Manoj K, Arumugam T, Prakash A. A comparative study on carbon sequestration potential of disturbed and undisturbed mangrove ecosystems in Kannur district, Kerala, South India. *Results Eng*. 2024;21:101716. <https://doi.org/10.1016/j.rineng.2023.101716>.
26. Patley MK, Tiwari A, Kumar K, Arumugam T, Kinattinkara S, Arumugam M. Study of mountain ecosystem accounting in lower Himalaya range in Uttarkhand, India using geospatial technology. *Results Eng*. 2024;21:101811. <https://doi.org/10.1016/j.rineng.2024.101811>.
27. Ganasri BP, Ramesh H. Assessment of soil erosion by RUSLE model using remote sensing and GIS—a case study of Nethravathi Basin. *Geosci Front*. 2016;7(6):953–61.
28. Yusof NF, Lihan T, Idris WMR, Rahman ZA, Mustapha MA, Yusof MAW. Prediction of soil erosion in Pansoon Sub-basin, Malaysia using RUSLE integrated in geographical information system. *Sains Malays*. 2019;48(11):2565–74. <https://doi.org/10.17576/jsm-2019-4811-26>.
29. Wagari M, Tamiru H. RUSLE model based annual soil loss quantification for soil erosion protection: a case of Fincha catchment. Ethiopia. 2021. <https://doi.org/10.1177/11786221211046234>.
30. Bonilla CA, Reyes JL, Magri A. Water erosion prediction using the Revised Universal Soil Loss Equation (RUSLE) in a GIS framework, central Chile. *Chil J Agricultural Res*. 2010;70(1):159–69.
31. Sidi Almouctar MA, Wu Y, Zhao F, Dossou JF. Soil erosion assessment using the RUSLE model and geospatial techniques (remote sensing and GIS) in South-Central Niger (Maradi Region). *Water*. 2021;13:3511.
32. Atoma H, Suryabagavan KV, Balakrishnan M. Soil erosion assessment using RUSLE model and GIS in Huluka watershed, Central Ethiopia. *Sustain Water Resour Manag*. 2020. <https://doi.org/10.1007/s40899-020-00365-z>.
33. Haokip P, Khan MA, Choudhari P, Kulimushi MC, Qaraev I. Identification of erosion-prone areas using morphometric parameters, land use land cover and multi-criteria decision-making method: geo-informatics approach. *Environ Dev Sustain*. 2022;24(1):527–57. <https://doi.org/10.1007/s10668-021-01452-7>.
34. Thangavelu Arumugam S, Kinattinkara K, Vellingiri M, Arumugam J, Rajamani A Jayaseelan. Assessment of agricultural soil quality in macro and micronutrient analysis of Kasargod, Kerala, India, using GIS techniques. *J Hazard Mater Adv*. 2025b;100846:2772–4166. <https://doi.org/10.1016/j.hazadv.2025.100846>.
35. Yousuf A, Sharma N, Bhat A, Khokhar M, Singh Sandhu A, Singh Dhillon P, Abdo B H. G. Soil erosion assessment and watershed prioritization using GIS-based RUSLE in lower Shivaliks of Northwest India. *Geol Ecol Landscapes*. 2025. <https://doi.org/10.1080/24749508.2025.2450108>.
36. Ashiagbor G, Forkuo EK, Laari P, Aabeyir R. Modeling soil erosion using RUSLE and GIS tools. *Int J Remote Sens Geosci*. 2013;2(4):1–17.
37. Ostovari Y, Ghorbani-Dashtaki S, Bahrami H-A, Naderi M, Dematte JAM. Soil loss estimation using RUSLE model, GIS and remote sensing techniques: a case study from the Dembecha Watershed, Northwestern Ethiopia. *Geoderma Reg*. 2017;11:28–36. <https://doi.org/10.1016/j.geodrs.2017.06.003>.

38. Panditharathne DL, Abeysingha NS, Nirmanee KG, Mallawatantri A. Application of revised universal soil loss equation (RUSLE) model to assess soil erosion in Kalu Ganga River Basin in Sri Lanka. *Appl Environ Soil Sci*. 2019. <https://doi.org/10.1155/2019/4037379>.
39. Kolli MK, Opp C, Groll M. Estimation of soil erosion and sediment yield concentration across the Kolleru Lake catchment using GIS. *Environ Earth Sci*. 2021;80(4):161. <https://doi.org/10.1007/s12665-021-09443-7>.
40. Attar MI, Pandey Y, Naseer S, et al. Soil erosion modelling using GIS-integrated RUSLE of Urpash watershed in Lesser Himalayas. *Arab J Geosci*. 2024;17:104. <https://doi.org/10.1007/s12517-024-11893-9>.
41. George JK, Kumar S, Hole RM. Geospatial modeling of soil erosion and risk assessment in Indian Himalaya region-A study of Uttarakhand state. *Environ Adv*. 2021;4:100039. <https://doi.org/10.1016/j.envadv.2021.100039>.
42. Jain MK, Das D. Estimation of sediment yield and areas of soil erosion and deposition for watershed prioritization using GIS and remote sensing. *Water Resour Manag*. 2010;24(10):2091–112.
43. Woldemariam G, Iguala A, Tekalign S, Reddy R. Spatial Modeling of Soil Erosion Risk and Its Implication for Conservation Planning: The Case of the Gobeles Watershed, East Hararghe Zone. *Ethiopia Land*. 2018;7:25.
44. Thangavelu A, Manoj K, Sapna K, Najla P. Monitoring the substantial metal analysis and HMPI in groundwater from village and nearby developed areas of kannur region: a GIS study. *Poll Res*. 2021b;40(4):1293–300.
45. Stefanidis S, Alexandridis V, Ghosal K. Assessment of water-induced soil erosion as a threat to Natura 2000 protected area in Crete Island, Greece. *Sustainability*. 2022;14:2738.
46. Alitane A, Essahlaoui A, El Hafyani M, El Hmadi A, El Ouali A, Kassou A, et al. Water erosion monitoring and prediction in response to the effects of climate change using RUSLE and SWAT equations: case of R'Dom watershed in Morocco. *Land*. 2022;11(1):93. <https://doi.org/10.3390/land11010093>.
47. Fiener P, Dost'ál T, Kr'asa J, Schmaltz E, Strauss P, Wilken F. Operational USLE-based modelling of soil erosion in Czech Republic, Austria, and Bavaria—differences in model adaptation, parametrization, and data availability. *Appl Sci*. 2020;10:3647.
48. Wynants M, Solomon H, Ndakidemi P, Blake WH. Pinpointing areas of increased soil erosion risk following land cover change in the Lake Manyara catchment, Tanzania. *Int J Appl Earth Obs Geoinf*. 2018;71:1–8.
49. Kinattinkara S, Arumugam T, Kuppusamy S, Krishnan M. Land use/land cover changes of Noyyal watershed in Coimbatore district, India, mapped using remote sensing techniques. *Environ Sci Pollut Res Int*. 2022. <https://doi.org/10.1007/s11356-022-18707-z>.
50. Zhang Z, Yu R. Assessment of soil erosion from an ungauged small watershed and its effect on Lake Ulansuhai, China. *Land*. 2023;12:440. <https://doi.org/10.3390/land12020440>.
51. Thangavelu Arumugam SK, Dewali, Snehmani SK. Estimation of snow depth in GIS environment from observation points on Z Gali region: A case study of NW Himalaya. *Results Eng*. 2024;21:101869. <https://doi.org/10.1016/j.rineng.2024.101869>.
52. Chen S, Zha X, Bai Y, Wang L. Evaluation of soil erosion vulnerability on the basis of exposure, sensitivity, and adaptive capacity: a case study in the Zhuxi watershed, Changting, Fujian Province, Southern China. *CATENA*. 2019;177:57–69.
53. Thangavelu Arumugam S, Kinattinkara K, Vellingiri, Anjana C. A GIS-Based Assessment of Groundwater Quality and Public Health Risks of Heavy Metals in Kannur, Kerala. *J Clean Water*. 2025c. <https://doi.org/10.1016/j.clwat.2025.100113>, 100113, ISSN 2950 – 2632.
54. Da Cunha ER, Bacani VM, Panachuki E. Modeling soil erosion using RUSLE and GIS in a watershed occupied by rural settlement in the Brazilian Cerrado. *Nat Hazards*. 2017;85(2):851–68.
55. Gupta S, Kumar S. Simulating climate change impact on soil erosion using RUSLE model – a case study in a watershed of mid-Himalayan landscape. *J Earth Syst Sci*. 2017;126:1–20. <https://doi.org/10.1007/s12040-017-0823-1>.
56. Farhan Y, Nawaiseh S. Spatial assessment of soil erosion risk using RUSLE and GIS techniques. *Environ Earth Sci*. 2015;74:4649–69.
57. Mahala A. Soil erosion estimation using RUSLE and GIS techniques—a study of a plateau fringe region of tropical environment. *Arab J Geosci*. 2018;11(13):335. <https://doi.org/10.1007/s12517-018-3703-3>.
58. Nampak H, Pradhan B, Mojaddadi Rizeei H, Park HJ. Assessment of land cover and land use change impact on soil loss in a tropical catchment by using multitemporal SPOT-5 satellite images and Revised Universal Soil Loss Equation model. *Land Degrad Dev*. 2018;29(10):3440–55. <https://doi.org/10.1002/ldr.3112>.
59. Djoukbal O, Mazour M, Hasbaia M, Benselama O. Estimating of water erosion in semiarid regions using RUSLE equation under GIS environment. *Environ Earth Sci*. 2018;77:345.
60. Mohammed S, Alsafadi K, Talukdar S, Kiwan S, Hennawi S, Alshihabi O, Harsanyie E. Estimation of soil erosion risk in southern part of Syria by using RUSLE integrating geo informatics approach. *Remote Sens Appl Soc Environ*. 2020;20:100375. <https://doi.org/10.1016/j.rsase.2020.100375>.
61. Burrough PA, McDonnell RA. *Principles of Geographical Information Systems*. New York, NY: Oxford University Press; 1998.
62. Wang L, Huang J, Du Y, Hu Y, Han P. Dynamic assessment of soil erosion risk using landsat TM and HJ Satellite Data in Danjiangkou Reservoir Area, China. *Remote Sens*. 2013;5:3826–48. <https://doi.org/10.3390/rs5083826>.
63. Thangavelu A, Sapna K, Sathiyaraj SR, Balasubramanian S. Spatial Temporal Analysis of rural crime hot spot zone using GIS: A part of Coimbatore. *Am J Geographical Res Reviews*. 2020;3:15. <https://doi.org/10.28933/ajgr-2019-08-1205>.
64. Tsegaye L, Bharti R. Soil erosion and sediment yield assessment using RUSLE and GIS-based approach in Anjeb watershed, Northwest Ethiopia. *SN Appl Sci*. 2021;3:582.
65. Thomas J, Joseph S, Thrivikramji KP. Assessment of soil erosion in a tropical mountain river basin of the southern Western Ghats, India using RUSLE and GIS. *Geosci Front*. 2018;9(3):893–906. <https://doi.org/10.1016/j.gsf.2017.05.011>.
66. Thangavelu Arumugam SK, Dewali, Snehmani and Ashwagoshia Ganju. Integration of terrain and AVHRR-derived multi-temporal snow cover data for statistical assessment of avalanches: Case study of a part of NW Himalaya. *Arab J Geosci*. 2019;12:539. <https://doi.org/10.1007/s12517-019-4691-7>.
67. Chalise D, Kumar L, Shrivastav CP, Lamichhane S. Spatial assessment of soil erosion in a hilly watershed of Western Nepal. *Environ Earth Sci*. 2018;77(19):685. <https://doi.org/10.1007/s12665-018-7842-3>.
68. Pandey A, Mathur A, Mishra SK, Mal BC. Soil erosion modelling of a Himalayan watershed using RS and GIS. *Environ Earth Sci*. 2009;59(2):399–410.
69. Ramzan S, Ganai MA, Ganie SA, Mir IA, Khan IA, Rouf AB, Ashraf I. Spatial variation characteristics of soil erodibility in the Ganderbal watershed of Sind River, Jammu and Kashmir (India). *Soil Water Conserv Bull*. 2020;5:17–26.

70. Wang R, Zhang S, Yang J, Pu L, Yang C, Yu L, Bu K. Integrated use of GCM, RS, and GIS for the assessment of hillslope and gully erosion in the Mushi River Sub-Catchment, Northeast China. *Sustainability*. 2016;8(4):317. <https://doi.org/10.3390/su8040317>.
71. Thakuriah G. GIS-based revised universal soil loss equation for estimating annual soil erosion: a case of lower Kulsi basin, India. *SN Appl Sci*. 2023;5:81. <https://doi.org/10.1007/s42452-023-05303-0>.
72. Maqbool S, Singh V, Patley MK, Kinattinkara S, Arumugam T. Evaluation of groundwater quality potential zones using AHP and WIOA models in Shopian District, Jammu and Kashmir, India: a GIS. *J Hazard Mater Adv*. 2024;16:100488. <https://doi.org/10.1016/j.hazadv.2024.100488>.
73. Shabani F, Kumar L, Esmaeili A. Improvement to the prediction of the USLE K factor. *Geomorphology*. 2014;204:229–34.
74. Sharma N, Kaushal A, Yousuf A, Kaur S, Sharda R. Prioritization of sub-watersheds and subsequent site identification for soil water and conservation practices using the SWAT-AHP integrated model in the Lower Sutlej sub-basin, India. *Environ Sci Pollut Res*. 2024;31(15):23120–45. <https://doi.org/10.1007/s11356-024-32382-2>.
75. Buraka T, Elias E, Suryabhagavan KV, Lelago A. Assessment of soil erosion risks in response to land-use and land-cover changes in Coka watershed, Southern Ethiopia. *Geol. Ecol Landscapes*. 2022. <https://doi.org/10.1080/24749508.2022.2109825>.
76. Negese A, Fekadu E, Getnet H. Potential Soil Loss Estimation and Erosion-Prone Area Prioritization Using RUSLE, GIS, and Remote Sensing in Chereti Watershed, Northeastern Ethiopia. *Air Soil Water Res*. 2021;14:1178622120985814.
77. Devonport BG, Evans C, Puig K, P. GIS-based rapid assessment of erosion risk in a small catchment in the wet/dry tropics of Australia. *Land Degrad Dev*. 2001;12:417–34.
78. Dabral PP, Baithuri N, Pandey A. Soil erosion assessment in a hilly catchment of North Eastern India using USLE, GIS and remote sensing. *Water Resour Manag*. 2008;22:1783–98.
79. Tian YC, Zhou YM, Wu BF, Zhou WF. Risk Assessment of Water Soil Erosion in Upper Basin of Miyun Reservoir. Volume 57. Beijing, China: Environmental Geology; 2009. p. 937e942.
80. Renard KG, Foster GR, Weesies GA, McCool DK, Yoder DC. Predicting soil erosion by water—a guide to conservation planning with the Revised Universal Soil Loss Equation (RUSLE). United States Department of Agriculture, Agricultural Research Service (USDA-ARS) Handbook No. 703. United States Government Printing Office: Washington, DC; 1997.
81. Angima SD, Stott DE, O'Neill MK, Ong CK, Weesies. GA. Soil erosion prediction using RUSLE for central Kenyan highland conditions. *Agric Ecosyst Environ*. 2003;97(1–3):295–308.
82. Wischmeier WH, Smith DD. Predicting rainfall erosion losses—a guide to conservation planning. *Agriculture Handbook No. 537*. Washington: US Department of Agriculture Science and Education Administration; 1978. p. 163.
83. Arnoldus HJM. An approximation of the rainfall factor in the Universal Soil Loss Equation. In: De Boodt M, Gabriels D, editors. *Assessment of Erosion*. Chichester, UK: Wiley; 1980. p. 127–32.
84. Van der Knijff JM, Jones RJA, Montanarella L. Soil erosion risk assessment in Europe. European Soil Bureau, EUR 19044 EN; 2000.
85. Durigon VL, Carvalho DF, Antunes MAH, Oliveira PTS, Fernandes MM. NDVI time series for monitoring RUSLE cover management factor in a tropical watershed. *Int J Remote Sens*. 2014;35(2):441–53.
86. Kouli M, Soupios P, Vallianatos F. Soil erosion prediction using the Revised Universal Soil Loss Equation (RUSLE) in a GIS framework, Chania, Northwestern Crete, Greece. *Environ Geol*. 2009;57:483–97.
87. Bhattacharya T, Sarkar D, Banerjee S. Estimation of soil erosion using RUSLE and GIS techniques: a case study of Eastern India. *Sustainable Water Resour Manage*. 2020;6(4):68.
88. Pandey A, Chowdary VM, Mal BC. Identification of critical erosion prone areas in the small agricultural watershed using USLE, GIS and remote sensing. *Water Resour Manage*. 2007;21(4):729–46.
89. Sarkar P, Sarma US, Gayen SK. Prioritization of sub-watersheds of Teesta River according to soil erosion susceptibility using multi-criteria decision-making in Sikkim and West Bengal. *Arab J Geosci*. 2023;16:398. <https://doi.org/10.1007/s12517-023-11423-z>.
90. Sarkar P, Gayen SK. Prioritization of sub-watersheds in the Lish tributary of the Teesta River Basin. *Discov Water*. 2025;5:60. <https://doi.org/10.1007/s43832-025-00262-6>.
91. Sarkar P, Gayen SK. Application of Entropy-AHP and WASPAS methods for prioritizing the sub-watersheds of Teesta River basin in terms of soil erosion susceptibility. *Discov Environ*. 2024;2:135. <https://doi.org/10.1007/s44274-024-00163-w>.
92. Sarkar P, Mondal M, Roy K, Sarma US, Gayen SK. Morphometric Analysis to Prioritize for Flood Risk of Sub-Watersheds of Teesta (Sikkim and West Bengal) Through Hazard Degree (HD) and Principal Component Analysis with Weighted Sum Approach (PCAWSA). In: Islam A, Das P, Ghosh S, Mukhopadhyay A, Gupta D, Singh AK, A, editors. *Fluvial Systems in the Anthropocene*. Cham: Springer; 2022. https://doi.org/10.1007/978-3-031-11181-5_13.
93. Das S, Jain MK, Auerswald K, et al. Identifying monthly rainfall erosivity patterns using hourly rainfall data across India. *Sci Rep*. 2025;15:27940. <https://doi.org/10.1038/s41598-025-11992-x>.
94. Majhi A, Shaw R, Mallick K, Patel PP. Towards improved USLE-based soil erosion modelling in India: a review of prevalent pitfalls and implementation of exemplar methods. *Earth Sci Rev*. 2021;221:103786.
95. Das S, Jain MK, Gupta V. An assessment of anticipated future changes in water erosion dynamics under climate and land use change scenarios in South Asia. *J Hydrol*. 2024;637:131341.
96. Maurya SK, Singh V, Chand K, Mishra PK, Abdelrahman K, Fnais MS, Withanage NC. Impact of Land Use Land Cover Dynamics and Climate Change on Snow Cover and Farmland Abandonment in the Upper Beas Valley, Western Himalaya, India. *Int J Inform Technol Decis Mak*. 2025. <https://doi.org/10.1142/S0219622025501068>.
97. Maurya SK, Singh V, Chand K, Mishra PK. Perceptions and Practices of Indigenous Soil Conservation Measures in Beas Valley, Himachal Pradesh, India. *GeoJournal*. 2024. <https://doi.org/10.1007/s10708-024-11068-9>.
98. Maurya SK, Singh V, Chand K, Mishra PK. Assessment of Soil Erosion in the Beas Valley, Kullu, Himachal Pradesh: A Study of Western Himalayan Landscape, Northern India. *Soil Sci Annual*. 2024. <https://doi.org/10.37501/soilsa/185558>.
99. Bagarello V, Di Stefano C, Ferro V, Giuseppe G, Iovino M. A pedotransfer function for estimating the soil erodibility factor in Sicily. *J Agric Eng*. 2009;40(3):7–13.
100. Gupta S, Borrelli P, Panagos P, Alewell C. An advanced global soil erodibility (K) assessment including the effects of saturated hydraulic conductivity. *Sci Total Environ*. 2024;908:168249. <https://doi.org/10.1016/j.scitotenv.2023.168249>.
101. Quijano L, Gaspar L, Navas A. Spatial patterns of SOC, SON, 137Cs and soil properties as affected by redistribution processes in a Mediterranean cultivated field (Central Ebro Basin). *Soil Tillage Res*. 2016;155:318–28.

102. Kebede M, Muhammad L, Rehman K, Tränckner J. Sediment yield estimation and evaluating the best management practices in the Nashe watershed, Blue Nile Basin, Ethiopia Environ. Monit Assess. 2023. <https://doi.org/10.1007/s10661-023-11337-z>.
103. Dahiphale P, Kumar M. Integrating RUSLE Model in Google Earth Engine and Geographic Information System for Soil Erosion Assessment—A Case Study of Upper Mahi River Basin, India. Eurasian Soil Sc. 2026;59:44. <https://doi.org/10.1134/S1064229325603531>.

Publisher's Note

Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.